Exercise 3.1: Properties of the Propagator The propagator $K(x_1, t_1; x_0, t_0)$ is defined as the transition amplitude between two space-time points, obtained by applying the time-evolution operator $U(t_1, t_0)$ between position eigenstates:

$$K(x_1, t_1; x_0, t_0) = \langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle . \tag{1}$$

We will prove a few properties of the propagator in th is exercise.

a) Show that the propagator obeys Schrödinger's equation by differentiating the definition of the propagator with respect to t_1 and using $\partial U(t_1, t_0)/\partial t_1 = -\frac{i}{\hbar}\hat{H}(t_1)U(t_1, t_0)$.

b) Show that the wavefunction at time t_1 can be expressed in terms of the propagator and the initial wavefunction at time t_0 as

$$\psi(x_1, t_1) = \int dx_0\, K(x_1, t_1; x_0, t_0)\psi(x_0, t_0), \tag{2}$$

where $\psi(x, t) = \langle x|\psi(t)\rangle$.

c) prove the following properties of the propagator:

- **Initial condition**: At equal times, the propagator reduces to the Dirac delta function

$$K(x_1, t; x_0, t) = \delta(x_1 - x_0). \tag{3}$$

- **Composition**: The propagator satisfies the composition property

$$K(x_2, t_2; x_0, t_0) = \int dx_1\, K(x_2, t_2; x_1, t_1)K(x_1, t_1; x_0, t_0), \tag{4}$$

for any intermediate time t_1.

d) Using the free-particle propagator

$$K(\mathbf{x}_1, t; \mathbf{x}_0, 0) = \left(\frac{m}{2\pi i\hbar t}\right)^{3/2} \exp\left(\frac{im|\mathbf{x}_1 - \mathbf{x}_0|^2}{2\hbar t}\right) \tag{5}$$

show that

$$K(\mathbf{p}_1, t; \mathbf{p}_0, 0) = \langle \mathbf{p}_1 | \hat{U}(t) | \mathbf{p}_0 \rangle = \delta(\mathbf{p}_1 - \mathbf{p}_0)e^{-\frac{i}{\hbar}E_{p_0}t}. \tag{6}$$

where $E_{p_0} = \frac{|\mathbf{p}_0|^2}{2m}$ is the kinetic energy of a free particle with momentum $\mathbf{p}_0$.

e) **(Bonus)** Starting from the time-evolution operator in the time domain (set $t_1 = t$ and $t_0 = 0$), derive the expression for the propagator in the energy representation (you can assume the Hamiltonian is time-independent):

$$K(E) = \frac{\hbar}{i}\frac{1}{E - \hat{H} + i\epsilon}, \tag{7}$$

where ϵ is a small positive number introduced to ensure convergence of the integral (we will return to this point in time-dependent perturbation theory).

Solutions to Exercise 3.1

a) We start from the definition of the propagator:

$$K(x_1, t_1; x_0, t_0) = \langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle \,. \tag{8}$$

Differentiating with respect to t_1, we have

$$\frac{\partial}{\partial t_1} K(x_1, t_1; x_0, t_0) = \langle x_1 | \frac{\partial \hat{U}(t_1, t_0)}{\partial t_1} | x_0 \rangle \,. \tag{9}$$

Using the relation $\partial U(t_1, t_0)/\partial t_1 = -\frac{i}{\hbar}\hat{H}(t_1) U(t_1, t_0)$, we get

$$\begin{aligned}
\frac{\partial}{\partial t_1} K(x_1, t_1; x_0, t_0) &= -\frac{i}{\hbar} \langle x_1 | \hat{H}(t_1)\hat{U}(t_1, t_0) | x_0 \rangle & (10)\\
&= \frac{-i}{\hbar} H(x, t_1) \langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle & (11)\\
&= \frac{-i}{\hbar} H(x, t_1) K(x_1, t_1; x_0, t_0). & (12)
\end{aligned}$$

which is Schrödinger's equation for the propagator.

Let me pause to clarify the step in Eq. (11). It is simply the statement that for a general ket $|\psi\rangle$:

$$\langle x | \hat{H}(t_1) | \psi \rangle = \langle x | \left(\hat{H}(t_1) | \psi \rangle \right) = H(x, t_1)\psi(x, t_1) = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x, t_1) \right) \psi(x, t_1), \tag{13}$$

where we interpreted $\hat{H}(t_1) |\psi\rangle$ as the new vector that we are going to project onto the position basis using $\langle x|$. In that sense, it is not that we are passing $\hat{H}(t_1)$ through $\langle x_1|$, but rather we are applying the Hamiltonian operator to the ket first, and then projecting onto position, which gives us the differential operator acting on the wavefunction. And to clarify, the operator $\hat{H}(t_1)$ acts on the Hilbert space of kets, and the "operator" ∇^2, for example, actts on the space of functions of position.
In other words, the equation:

$$\langle x_1 | \hat{H}(t_1) | \psi \rangle = H(x_1, t_1)\psi(x_1, t_1), \tag{14}$$

is telling you how to turn the operation on the Hilbert space into the operation of $H(x_1, t_1)$ on the space of functions of position. In the same way, the equation:

$$\langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle = K(x_1, t_1; x_0, t_0), \tag{15}$$

tells you how to turn the operation on the Hilbert space into the operation of $K(x_1, t_1; x_0, t_0)$ on the space of functions of position (in this case, a trivial multiplication by a complex exponential function).

b) The wavefunction at time t_1 is given by

$$\psi(x_1, t_1) = \langle x_1 | \hat{U}(t_1, t_0) | \psi(t_0) \rangle \,. \tag{16}$$

Inserting a complete set of position eigenstates at time t_0, we have

$$\psi(x_1, t_1) = \int dx_0 \, \langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle \langle x_0 | \psi(t_0) \rangle \,. \tag{17}$$

Recognizing the first term as the propagator and the second term as the initial wavefunction, we obtain

$$\psi(x_1, t_1) = \int dx_0 K(x_1, t_1; x_0, t_0)\psi(x_0, t_0). \tag{18}$$

c) Initial condition: At equal times, the time-evolution operator reduces to the identity operator, $\hat{U}(t,t) = \mathbb{1}$. Thus, the propagator becomes

$$K(x_1, t; x_0, t) = \langle x_1 | \mathbb{1} | x_0 \rangle = \langle x_1 | x_0 \rangle = \delta(x_1 - x_0). \tag{19}$$

Composition: We start from the definition of the propagator:

$$K(x_2, t_2; x_0, t_0) = \langle x_2 | \hat{U}(t_2, t_0) | x_0 \rangle = \langle x_2 | \hat{U}(t_2, t_1)\hat{U}(t_1, t_0) | x_0 \rangle , \tag{20}$$

where we are using the composition property of the time-evolution operator (this one is trivial to prove as well). Inserting a complete set of position eigenstates at an intermediate time t_1, we have

$$K(x_2, t_2; x_0, t_0) = \int dx_1 \, \langle x_2 | \hat{U}(t_2, t_1) | x_1 \rangle \langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle . \tag{21}$$

Recognizing the terms as propagators, we obtain

$$K(x_2, t_2; x_0, t_0) = \int dx_1 K(x_2, t_2; x_1, t_1) K(x_1, t_1; x_0, t_0). \tag{22}$$

d) To Fourier transform to momentum space, we use the definition of $K(\mathbf{p}_1, t_1; \mathbf{p}_0, t_0)$ and insert complete sets of position eigenstates:

$$K(\mathbf{p}_1, t_1; \mathbf{p}_0, t_0) \equiv \langle \mathbf{p}_1 | \hat{U}(t) | \mathbf{p}_0 \rangle = \int d\mathbf{x}_1 \int d\mathbf{x}_0 \, \langle \mathbf{p}_1 | \mathbf{x}_1 \rangle \, K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) \, \langle \mathbf{x}_0 | \mathbf{p}_0 \rangle . \tag{23}$$

Using the plane wave representation of momentum eigenstates in position space, we have

$$\langle \mathbf{p} | \mathbf{x} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{-\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{x}}. \tag{24}$$

Substituting this into the expression for the momentum space propagator, we get

$$K(\mathbf{p}_1, t; \mathbf{p}_0, 0) = \frac{1}{(2\pi\hbar)^3} \int d\mathbf{x}_1 \int d\mathbf{x}_0 e^{-\frac{i}{\hbar}\mathbf{p}_1\cdot\mathbf{x}_1} K(\mathbf{x}_1, t; \mathbf{x}_0, 0) e^{\frac{i}{\hbar}\mathbf{p}_0\cdot\mathbf{x}_0}. \tag{25}$$

Substituting the free-particle propagator into the integral, we have

$$K(\mathbf{p}_1, t; \mathbf{p}_0, 0) = \frac{1}{(2\pi\hbar)^3} \left(\frac{m}{2\pi i\hbar t}\right)^{3/2} \int d\mathbf{x}_1 \int d\mathbf{x}_0 e^{-\frac{i}{\hbar}\mathbf{p}_1\cdot\mathbf{x}_1} \exp\left(\frac{im|\mathbf{x}_1 - \mathbf{x}_0|^2}{2\hbar t}\right) e^{\frac{i}{\hbar}\mathbf{p}_0\cdot\mathbf{x}_0}. \tag{26}$$

To evaluate the integrals, we complete the square in the exponent. After some algebra, we find that the integrals yield delta functions, leading to

$$K(\mathbf{p}_1, t; \mathbf{p}_0, 0) = \delta(\mathbf{p}_1 - \mathbf{p}_0) e^{-\frac{i}{\hbar}E_{p_0}t}, \tag{27}$$

where $E_{p_0} = \frac{|\mathbf{p}_0|^2}{2m}$ is the kinetic energy of a free particle with momentum $\mathbf{p}_0$. This shows that in momentum space, the free particle propagator simply adds a phase factor to each momentum eigenstate, as expected.

e) (Bonus) Starting from the time-evolution operator (assuming a time-independent Hamiltonian),

$$\hat{U}(t_1, t_0) = e^{-\frac{i}{\hbar}\hat{H}t}, \tag{28}$$

the propagator in the energy representation is given by the Fourier transform of the time-evolution operator:

$$K(E) = \int_0^\infty dt\, e^{\frac{i}{\hbar}(E)t}\hat{U}(t) = \int_0^\infty dt\, e^{\frac{i}{\hbar}(E)t}e^{-\frac{i}{\hbar}\hat{H}t}. \tag{29}$$

Combining the exponentials, we have

$$K(E) = \int_0^\infty dt\, e^{\frac{i}{\hbar}(E-\hat{H})t}. \tag{30}$$

To ensure convergence of the integral, we introduce a small positive imaginary part to the energy, i.e., we replace E with $E + i\epsilon$ where $\epsilon > 0$. Thus, we have

$$K(E) = \int_0^\infty dt\, e^{\frac{i}{\hbar}(E-\hat{H}+i\epsilon)t}. \tag{31}$$

Evaluating the integral, we find

$$K(E) = \left[\frac{\hbar}{i(E-\hat{H}+i\epsilon)}e^{\frac{i}{\hbar}(E-\hat{H}+i\epsilon)t}\right]_0^\infty. \tag{32}$$

Taking the limit as $t \to \infty$, the exponential term vanishes due to the positive imaginary part, leaving us with

$$K(E) = \frac{\hbar}{i}\frac{1}{E-\hat{H}+i\epsilon}. \tag{33}$$

This is a little more subtle to interpret than the propagator in the time domain, but the poles of this function in the complex energy plane correspond to the energy eigenvalues of the Hamiltonian, and the $i\epsilon$ prescription ensures that we are considering the retarded Green's function, which is causal. We may prove this later, but for now, you can interpret this result as a tool to study the energy spectrum of the system.